

The damped three-point cosine min–max constant: a certified identification

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Abstract

For $z_j = r_j e^{i\varphi_j}$ with $|z_j| \leq 1$ and $\max_j |z_j| = 1$, consider the *short-window* real-part power sum $\max_{1 \leq k \leq 5M} \operatorname{Re} \sum_{j \leq M} z_j^k$. For $M = 3$ the worst configuration is *damped*, i.e. it lies in the interior of the polydisc. We determine the corresponding constant exactly:

$$C_3 = 0.373\,091\,892\,895\,816\,4\dots$$

and show that, in the domain of definition $\Omega = [0, 1]^2 \times [0, \pi]^3$, the minimiser set consists of exactly two points, z_* and its partner σz_* under the paired swap of the two damped factors. The proof is a three-piece splice: a certified interval Krawczyk enclosure of a six-branch nonsmooth KKT root (existence, uniqueness, strict positivity of all multipliers, and exact tangency at the root); a strict local-growth lemma centred at that root; and a certified global branch-and-bound exclusion on the complement of two balls. All computer-assisted statements carry an explicit interval certificate and a structurally independent reproduction.

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1 Setting and statement of the theorem

1.1 Definitions

Throughout, $M = 3$, $K := 5M = 15$, and the *domain* is the five-dimensional box

$$\Omega := [0, 1]^2 \times [0, \pi]^3 \subset \mathbb{R}^5, \quad x = (r_2, r_3, \varphi_1, \varphi_2, \varphi_3). \quad (1)$$

For $1 \leq k \leq K$ define

$$S_k(x) := \cos(k\varphi_1) + r_2^k \cos(k\varphi_2) + r_3^k \cos(k\varphi_3), \quad F(x) := \max_{1 \leq k \leq K} S_k(x), \quad C_3 := \inf_{x \in \Omega} F(x). \quad (2)$$

The *active set* at x is $A(x) := \{k \leq K : S_k(x) = F(x)\}$, and the *active spread* is

$$\delta_A(x) := F(x) - \min_{k \in A(x)} S_k(x) \geq 0. \quad (3)$$

The *paired swap* is the involution

$$\sigma : (r_2, r_3, \varphi_1, \varphi_2, \varphi_3) \mapsto (r_3, r_2, \varphi_1, \varphi_3, \varphi_2), \quad C_2 := \{\text{id}, \sigma\}. \quad (4)$$

Provenance of the parametrisation. Writing $z_j = r_j e^{i\varphi_j}$ one has $\text{Re } z_j^k = r_j^k \cos(k\varphi_j)$. Normalising the largest modulus to 1 and placing it first yields $r_1 = 1$, so the three-point, linear-window problem $\max_{1 \leq k \leq 5M} \text{Re } \sum_{j \leq M} z_j^k$ reduces to the study of F on Ω . Since $\cos$ is even and 2π -periodic, the angle reductions $\varphi_j \mapsto -\varphi_j$ and $\varphi_j \mapsto 2\pi - \varphi_j$ make $[0, \pi]$ a fundamental domain in each angle coordinate; no generality is lost.

The certified box. $X_0 \subset \mathbb{R}^{11}$ denotes the interval Krawczyk box produced in Lemma 3 (seed radius $r_0 = 10^{-12}$, final coordinate width $\leq 10^{-21}$), $\text{hd}(X_0)$ its half-diagonal, and z_* the unique point of Ω solving the KKT system of Lemma 3 inside X_0 . We write c_0 for the decimal centre of X_0 , used as the numerical centre of the exclusion balls in Lemma 6. Finally, $B(c, \rho)$ denotes the closed Euclidean ball in $\mathbb{R}^5$, $\text{far}(X, c) := \max_{y \in X} \|y - c\|$ (attained at a vertex of X), and $\underline{F}_{\text{IA}}(X)$ an interval-arithmetic lower bound for $\min_X F$.

1.2 Main theorem

Theorem 1 (Global minimisation for damped $M = 3$). *With Ω, F, C_3 as above,*

$$C_3 = F(z_*) = 0.373\,091\,892\,895\,816\,4\dots$$

and the minimiser set in the normalised domain is

$$\arg \min_{x \in \Omega} F(x) = \{z_*, \sigma z_*\},$$

two distinct points of Ω .

The proof is given in §7; it is the splice of Lemma 3 (the certified KKT root), Lemma 3 (active/non-active separation), Lemma 4 (a uniform first-order covering estimate), Lemma 5 (strict local growth at the root) and Lemma 6 (certified global exclusion).

Remark 2 (Minimiser set: normalised domain versus raw permutations). The minimiser set *in the normalised domain* Ω consists of the two points z_* and σz_* . The corresponding *raw* permutation orbit contains six configurations, whose images under the normalisation map $\mathcal{N}$ reduce to these two points, i.e. $|\mathcal{O}_{\text{raw}}| = 6$ while $|\mathcal{N}(\mathcal{O}_{\text{raw}})| = 2$ (Proposition 13). Statements about the number of minimisers always refer to Ω .

2 Lemma 1: certified KKT root and active-set positivity [CA]

Let $A_* := \{1, 2, 3, 4, 5, 15\}$ and consider the system in the eleven unknowns

$$z = (r_2, r_3, \varphi_1, \varphi_2, \varphi_3, \lambda_1, \lambda_2, \lambda_3, \lambda_4, \lambda_5, \lambda_{15}),$$

$$G(z) = \begin{pmatrix} \sum_{k \in A_*} \lambda_k \nabla S_k \\ S_1 - S_2 \\ S_1 - S_3 \\ S_1 - S_4 \\ S_1 - S_5 \\ S_1 - S_{15} \\ \sum_{k \in A_*} \lambda_k - 1 \end{pmatrix} = 0 \quad (5 + 5 + 1 = 11 \text{ equations}). \quad (5)$$

Lemma 3. *There is exactly one solution z_* of (5) with coordinate vector in the box X_0 . At that solution*

$$\lambda_{\min} = \min_{k \in A_*} \lambda_k = 0.111\,860\,12 > 0, \quad \|I - YJ(X_0)\|_{\infty} = 1.965\,1 \times 10^{-10} < 1,$$

and every coordinate width of X_0 is $\leq 10^{-21}$. Consequently the exact root satisfies the six tangency equations, i.e. $S_k(z_) = F(z_*)$ for all $k \in A_*$, and hence*

$$\delta_A(z_*) = 0. \quad (6)$$

Certification. The system (5) is solved by an interval Krawczyk operator

$$K(X) = m - YG(m) + (I - YJ(X))(X - m), \quad m = \text{mid}(X),$$

where J is the analytic Jacobian and Y an approximate inverse of J at the high-precision root. Two *separate* operator conditions are verified in interval arithmetic:

$$K(X) \subset \text{int } X \quad (\text{existence of a solution in } X); \quad \|I - YJ(X)\|_{\infty} < 1 \quad (\text{uniqueness}).$$

Existence and uniqueness are stated separately for this reason. The seed radius is $r_0 = 10^{-12}$; the system is ill-conditioned ($\text{cond} \approx 2 \times 10^4$, multipliers spanning 0.112–0.816, frequencies reaching 15) and with $r_0 = 10^{-6}$ one gets $\|I - YJ\| \approx 6 > 1$, so the contraction test fails — the small seed is necessary, not cosmetic. The six multipliers certified in X_0 are

$$(\lambda_1, \lambda_2, \lambda_3, \lambda_4, \lambda_5, \lambda_{15}) = (0.200\,054, 0.178\,488, 0.165\,510, 0.194\,819, 0.111\,860, 0.149\,270),$$

all strictly positive. Since the solution of (5) inside X_0 is unique and the interval enclosures of the five tie differences $S_1 - S_k$ ($k = 2, 3, 4, 5, 15$) all contain 0, the unique root satisfies the ties exactly; together with $\sum_k \lambda_k = 1$ and $\lambda_k > 0$ this gives $S_k(z_*) = F(z_*)$ for every $k \in A_*$, whence (6). $\square$

Remark 4. Containment of 0 in the interval enclosure of a tie *over the whole box* does not assert that the ties hold identically on X_0 . Only the equality at the unique root, (6), is used below.

3 Lemma 2: active/non-active separation [CA]

Lemma 5. *On the reference box X_{ref} (the variable part of X_0),*

$$\Delta_{\text{ref}} := \min_{k \in A_*} \inf_{X_{\text{ref}}} S_k - \max_{k \notin A_*} \sup_{X_{\text{ref}}} S_k = 0.279\,596\,813\,6 > 0. \quad (7)$$

With the Lipschitz constants of the two families on Ω ,

$$L_{\text{act}} := \max_{k \in A_*} k\sqrt{5} = 15\sqrt{5} = 33.541\,02, \quad L_{\text{non}} := \max_{k \notin A_*} k\sqrt{5} = 14\sqrt{5} = 31.304\,952,$$

the active set is constant on the ball $B(c_0, \rho_{\text{iso}})$ with

$$\rho_{\text{iso}} = \frac{\Delta_{\text{ref}}}{L_{\text{act}} + L_{\text{non}}} = 4.311\,71 \times 10^{-3}. \quad (8)$$

Certification. For a box X , $\inf_X S_k$ and $\sup_X S_k$ are evaluated *exactly per coordinate*: the extrema of $\cos(k\varphi)$ on an interval are attained at an endpoint or at an odd (resp. even) multiple of π inside it, and the damping factor r^k enters as an *interval product*. The Lipschitz bound $\|\nabla S_k\| \leq k\sqrt{5}$ holds because each of the five partial derivatives is bounded by k (displayed: $\partial_{r_j}(r_j^k \cos k\varphi_j) = kr_j^{k-1} \cos k\varphi_j$ and $\partial_{\varphi_j}(r_j^k \cos k\varphi_j) = -kr_j^k \sin k\varphi_j$, all $\leq k$ in modulus). Separation (7) then forces the active set to be constant on any ball of radius at most $\Delta_{\text{ref}}/(L_{\text{act}} + L_{\text{non}})$, giving (8). $\square$

Remark 6 (Why the interval product is indispensable). Bounding $r^k \cos(k\varphi)$ as “a positive factor times a cosine” by treating the two factors independently gives the spurious value $\Delta_{\text{ref}} = -0.522 < 0$, i.e. a false failure. The genuine product bound yields (7).

4 Lemma 3: uniform first-order covering estimate [CA]

Lemma 7. *At the certified root z_* the origin lies in the interior of the convex hull of the active gradients, and, after accounting for the data ball $B(c_0, \rho_{\text{data}})$ with $\rho_{\text{data}} = 2.05 \times 10^{-3}$, the covering constant*

$$c_X = c - \varepsilon_{\text{pert}} = 0.302\,091\,5 - 0.203\,2 = 0.098\,812\,009\,77 > 0 \quad (9)$$

satisfies

$$\max_{k \in A_*} \langle \nabla S_k(x), u \rangle \geq c_X \quad \text{for all } x \in B(c_0, \rho_{\text{data}}) \text{ and all } \|u\| = 1.$$

Certification. The six points $\{\nabla S_k(z_*)\}_{k \in A_*} \subset \mathbb{R}^5$ span a simplex. The quantity $c := \min_{\|u\|=1} \max_{k \in A_*} \langle \nabla S_k(z_*), u \rangle$ equals the largest ball about the origin contained in that simplex, i.e. the minimum, over its six facets, of the distance from the origin to the facet hyperplane. The facet normals are obtained from the null space of the corresponding 5×5 facet matrix, and a containment certificate (all remaining vertices strictly on one side of each facet) verifies that the inscribed ball is indeed interior. Propagating the enclosing data ball of radius ρ_{data} by the maximal gradient variation $\varepsilon_{\text{pert}} = 0.203\,2$ yields the usable constant c_X in (9). $\square$

5 Lemma 4: strict local growth at the certified root [CA]

Lemma 8. *Let*

$$R := \max_{k \in A_*} \sup_{x \in B(c_0, \rho_{\text{data}})} \|\nabla^2 S_k(x)\|_2 = 99.899\,695\,515\,930\,13, \quad \rho_g := \frac{2c_X}{R} = 1.978\,224\,4 \times 10^{-3}. \quad (10)$$

Then for every δ with $0 < \|\delta\| \leq \rho_g$,

$$\boxed{F(z_* + \delta) \geq F(z_*) + c_X \|\delta\| - \frac{R}{2} \|\delta\|^2 > F(z_*)} \quad (11)$$

Proof. Fix $x \in B(c_0, \rho_{\text{data}})$. By Taylor's theorem and (10),

$$S_k(x + \delta) \geq S_k(x) + \langle \nabla S_k(x), \delta \rangle - \frac{R}{2} \|\delta\|^2 \quad (k \in A_*).$$

Taking the maximum over $k \in A_*$ and using the elementary fact

$$\max_k (a_k + b_k) \geq \min_k a_k + \max_k b_k \quad (12)$$

(valid for arbitrary finite families; note that max is *subadditive*, so the reverse inequality is false) we obtain the *spread form*

$$\boxed{F(x + \delta) \geq F(x) - \delta_A(x) + c_X \|\delta\| - \frac{R}{2} \|\delta\|^2} \quad (13)$$

where $\delta_A(x)$ is the active spread (3). Centred at the root $x = z_*$, the identity (6) of Lemma 3 gives $\delta_A(z_*) = 0$, and (13) becomes (11). Finally $\rho_g = 2c_X/R$ is precisely the (smaller) root of $c_X \rho - \frac{R}{2} \rho^2 = 0$, so the right-hand side of (11) is strictly positive for all $0 < \|\delta\| < \rho_g$ and equals $F(z_*)$ only in the limit $\|\delta\| \rightarrow \rho_g$. $\square$

Remark 9 (The spread term is the true logical consumer of the tangency). If (13) were available with $\min_{k \in A_*} S_k(x)$ at an *arbitrary* point x of the reference box, then every such x would be a strict local minimiser of F , which is impossible. The honest statement therefore carries the spread $\delta_A(x)$, and Lemma 3 is consumed at exactly one place: the identity $\delta_A(z_*) = 0$.

Remark 10 (Self-consistency of the reference ball). The constants c_X and R must be valid on the whole ball on which (11) is asserted. This is a *proof condition*, not an implementation detail:

$$\boxed{\rho_{\text{data}} \geq \rho_g + \text{hd}(X_0)} \quad (14)$$

Indeed c_X is obtained by propagating the data ball $B(c_0, \rho_{\text{data}})$, while (11) is applied on $B(z_*, \rho_g)$; since $\|z_* - c_0\| \leq \text{hd}(X_0)$, (14) is exactly what makes the application legitimate. Here $\rho_{\text{data}} = 2.05 \times 10^{-3} \geq 1.978\,224\,4 \times 10^{-3} + 10^{-21}$. A configuration with $\rho_{\text{data}} = 10^{-5}$ and certified radius 4.3×10^{-3} violates (14) and was discarded.

6 Lemma 5: certified global exclusion [CA]

Lemma 11. Let ρ_g be as in (10), put $B_1 := B(z_*, \rho_g)$, $B_2 := B(\sigma z_*, \rho_g)$, and

$$T_C := \sup_{X_0} F + 10^{-9} = F(z_*) + 10^{-9} = 0.373\,091\,893\,895\,817. \quad (15)$$

Then $F(x) \geq T_C$ for every $x \in \Omega \setminus (B_1 \cup B_2)$.

Certification. Run an adaptive branch-and-bound on Ω with the exact separable box lower bound

$$\text{LB}(X) = \max_{1 \leq k \leq K} \left[\min_{I_1} \cos k\varphi_1 + \min_{I_2} (r_2^k \cos k\varphi_2) + \min_{I_3} (r_3^k \cos k\varphi_3) \right] \leq \min_X F,$$

the damped terms again evaluated as interval products. A box X is *discarded* if $\text{far}(X, c_0) \leq \rho_g - 10^{-9}$ from one of the two centres — then $X \subseteq B(c_0, \rho_g - 10^{-9}) \subseteq B(z_*, \rho_g)$ because $\text{hd}(X_0) \leq 10^{-21}$, so X lies in the region certified by Lemma 8; *certified* if $\underline{F}_{\text{IA}}(X) \geq T_C$; and *split* (bisection along the widest coordinate) otherwise, unless the box is already below the minimum width 10^{-7} , in which case it is counted *unresolved* and would invalidate the lemma. The outcome is:

boxes evaluated	N_{eval}	472 766
interval-certified	N_{cert}	252 282
discarded	N_{disc}	485
split	N_{split}	219 999
unresolved	N_{unres}	0
float-pass / interval-fail		0
$\min_X (\underline{F}_{\text{IA}}(X) - T_C)$		$4.628\,931 \times 10^{-8}$
$\max\{\text{far}(X, c_0) : X \text{ discarded}\}$		$0.001\,978\,039\,980 \leq \rho_g - 10^{-9}$

The adaptive segmentation is a disjoint tiling of Ω , and $N_{\text{unres}} = 0$ means every terminal box is either certified or discarded. Writing $\mathcal{C}_{\text{cert}}$ for the union of the certified boxes and $\mathcal{D}$ for the union of the discarded ones,

$$\boxed{\Omega = \mathcal{C}_{\text{cert}} \sqcup \mathcal{D}, \quad \mathcal{D} \subseteq B_1 \cup B_2, \quad F \geq T_C \text{ on } \mathcal{C}_{\text{cert}}} \quad (16)$$

Consequently $\Omega = \mathcal{C}_{\text{cert}} \cup B_1 \cup B_2$ is a valid cover: the first decomposition of (16) is exact, so no gap is possible, and overlaps among the three pieces are harmless. $\square$

7 Proof of Theorem 1

Let $x \in \Omega$. By Lemma 11 and (16), $\Omega = \mathcal{C}_{\text{cert}} \cup B_1 \cup B_2$.

1. If $x \in \mathcal{C}_{\text{cert}}$, then $F(x) \geq T_C = F(z_*) + 10^{-9} > F(z_*)$.
2. If $x \in B_1 \cup B_2$, write $x = \sigma^i z_* + \delta$ with $i \in \{0, 1\}$ and $\|\delta\| \leq \rho_g$. If $\delta = 0$, then $F(x) = F(z_*)$ by $F \circ \sigma = F$ (Lemma 12). Otherwise Lemma 8 applied at the corresponding centre gives $F(x) > F(z_*)$.

The two cases cover Ω , so $F(x) \geq F(z_*)$ on Ω with equality exactly at z_* and σz_* ; hence $C_3 = F(z_*)$ and the minimiser set in Ω is exactly $\{z_*, \sigma z_*\}$. The two points are distinct because $r_2 = 0.7905 \neq 0.8302 = r_3$. $\square$

The hypotheses consumed are: (6) from Lemma 3; the constants of Lemmas 5–8; and the target (15) with the partition outcome of Lemma 11. No property of the Riemann zeta function is used.

8 The minimiser set in the normalised domain

Lemma 12 (Symmetry). $F \circ \sigma = F$, where σ is the paired swap (4).

Proof. Applying σ to x merely exchanges the second and third summands of every S_k , and the maximum over k is unchanged. $\square$

Proposition 13. *Let z_* be the root of Lemma 3. Then*

- (i) *the raw permutation orbit of z_* under S_3 has six points, $|\mathcal{O}_{\text{raw}}| = 6$;*
- (ii) *under the normalisation map $\mathcal{N}$ (re-order the pairs so that the unit-modulus pair comes first) the orbit collapses to exactly two points,*

$$|\mathcal{N}(\mathcal{O}_{\text{raw}})| = 2, \quad \mathcal{N}(\mathcal{O}_{\text{raw}}) = \{z_*, \sigma z_*\};$$

- (iii) *permuting the three angles while keeping the radii fixed is not a symmetry of F : for the certified root,*

$$F(r_2, r_3, \varphi_2, \varphi_1, \varphi_3) = 1.126\,876\,598\,534\,850 \neq 0.373\,091\,892\,895\,817 = F(z_*).$$

Proof. (i) The three pairs $(1, \varphi_1), (r_2, \varphi_2), (r_3, \varphi_3)$ are pairwise distinct because $r_2 \neq r_3$; permuting them gives six distinct raw configurations, and the underlying functional $\max_k \operatorname{Re} \sum_j z_j^k$ is permutation invariant, so all six carry the same value. (ii) Of the six permutations, four move the unit-modulus pair into position 2 or 3. Re-normalising (moving the unit pair back to the first position and keeping the relative order of the remaining two) identifies those four with either id or σ according to whether the relative order is preserved or reversed. Hence at most two distinct normalised points, and they are distinct since $r_2 \neq r_3$. (iii) Interchanging φ_1 and φ_2 alone merges the undamped term $\cos k\varphi_1$ with the damped term $r_2^k \cos k\varphi_2$, which is not an invariance; the numerical values differ as stated. $\square$

9 Independent computational reproduction

The interval layer of Lemma 11 was implemented twice. The second implementation shares only the (deterministic) floating-point partition and differs structurally: the interval verification is distributed over three processes with an independently written interval class. It reproduces

$$N_{\text{cert}} = 252\,282, \quad \text{interval failures} = 0, \quad \min_X (\underline{F}_{\text{IA}}(X) - T_C) = 4.628\,931\,107\,212\,973\,4 \times 10^{-8},$$

matching the primary implementation to sixteen significant digits. The two implementations are recorded as scripts `dc2iv_strict.py` (sequential) and `dc2par_parallel_iv.py` (parallel).

10 Implementation caveats

1. **Facet normals via floating-point SVD.** Lemma 7 obtains the facet normals from a floating-point SVD. The perturbation allowance in the data ball is $\varepsilon_{\text{pert}} \approx 0.203$, so $10^{-15} \ll \varepsilon_{\text{pert}}$ and the floating-point error cannot change the sign of c_X . This is a numerical-implementation caveat: the lemma is *not* formalised in interval linear algebra.
2. **Self-built interval arithmetic.** Lemma 11 uses a hand-rolled interval class over `mpmath` at 40 decimal digits, with an outward slack of 10^{-30} per operation and π enclosed by rational endpoints. An alternative line using `mpmath.iv` reproduces the bounds independently.
3. **`mpmath` correctness** is assumed, as elsewhere in this project. The interval layer also reports the float-pass / interval-fail count, which is 0.
4. **Lipschitz constants.** The bounds $k\sqrt{5}$ in Lemma 5 are analytic, not computed.

5. R in (10) is an explicit conservative bound obtained from the 5×5 Hessians with $|\cos|, |\sin| \leq 1$ and the damping factors retained; since it is an upper bound, (11) is conservative.
6. **Size of the exclusion radius.** $\rho_g = 2c_X/R$ is covering-limited (it is smaller than the isolation radius ρ_{iso} of (8)); the growth inequality (11) is sharp only in its linear regime, and the margin at $\|\delta\| = \rho_g$ vanishes in the limit.

11 Boundaries: what is not claimed

1. Theorem 1 concerns $M = 3$ in the damped setting. No extrapolation to general M , to damped $M \geq 4$, or to the undamped constant m_M is claimed.
2. The value of C_3 is known at interval level; the displayed decimal is the upper-bound configuration's value together with the certified threshold $T_C = F(z_*) + 10^{-9}$, not a closed form.
3. No property of the Riemann zeta function is used.
4. Earlier brackets $(0.373\,091\,8 \leq C_3 \leq 0.373\,092\,075\,762$ and $0.76 \leq m_3 \leq 0.764\,081\,100\,745\,854)$ are superseded and retained only as checkpoints.
5. The proof is computer-assisted: Lemmas 3–11 are [CA] statements. The elementary layer — Lemma 12, Proposition 13, inequality (12), the Lipschitz bounds and the Taylor step — is proved by hand in this note.

A Numerical certificate table

item	value
certified root z_* (φ_j/π)	(0.10911016482862308881, 0.8206637095202066754, 0.4617193710186102)
radii	$r_2 = 0.79051323395036623846$, $r_3 = 0.83020729481457293027$
Krawczyk seed / width / $\ I - YJ\ _\infty$	10^{-12} / 3.93×10^{-22} / 1.9651×10^{-10}
$\lambda_{\min}$ ($k = 1, 2, 3, 4, 5, 15$)	0.111 860 12
$\Delta_{\text{ref}}, \rho_{\text{iso}}$	0.279 596 813 6, $4.311\,71 \times 10^{-3}$
$c, c_X, \rho_{\text{data}}$	0.302 091 5, 0.098 812 009 77, 2.05×10^{-3}
R, ρ_g	99.899 695 515 930 13, $1.978\,224\,4 \times 10^{-3}$
$T_C - F(z_*)$	10^{-9}
$N_{\text{eval}}, N_{\text{cert}}, N_{\text{disc}}, N_{\text{split}}, N_{\text{unres}}$	472 766, 252 282, 485, 219 999, 0
$\min_X(\underline{F}_{\text{IA}}(X) - T_C)$	$4.628\,931 \times 10^{-8}$
cross-check	identical to 16 significant digits
C_3	0.373 091 892 895 816 4...

References

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